

Question 9 – Investigating the Photoelectric Effect

(a)(i) Calculate the energy of a photon of the UV radiation

✓ 3 marks

Given:

- Frequency $f = 1.25 \times 10^{15}$ Hz
- Planck's constant $h = 6.63 \times 10^{-34}$ Js

Use:

$$E = hf = 6.63 \times 10^{-34} \times 1.25 \times 10^{15} = 8.29 \times 10^{-19} \text{ J}$$

(a)(ii) Explain why a current is produced

✓ 2 marks

Given:

- Work function of zinc $\phi = 5.81 \times 10^{-19}$ J
- Photon energy $E = 8.29 \times 10^{-19}$ J

✓ **Explanation:**

- The energy of the photons is **greater than the work function** of zinc, so **photoelectrons are emitted**
- These electrons are attracted to the **positive wire mesh**, allowing a **current to flow**

(b) Lamp moved closer to quartz window

✓ 2 marks

✓ **Answer:**

The **current increases**,
because **more photons strike the zinc plate per second**, causing **more photoelectrons to be emitted**

 *Justification required for full marks*

(c) Polarity of DC supply is reversed

✓ 2 marks

✓ **Answer:**

The **current becomes zero**
because the **photoelectrons are repelled by the now-negative mesh** and **cannot complete the circuit**

 **Acceptable alternatives:**

- Fewer or no photoelectrons reach the mesh
- The zinc plate becomes positively charged and attracts them back

 Revision Tips

- The photoelectric effect occurs when:

Photon energy $>$ Work function

- Increasing **intensity** increases **current**, not electron energy
- Reversing the polarity can **stop current** even if photoelectrons are emitted

